FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation

Anonymous Author(s)

Abstract

Federated recommendation (FR) enables collaborative model training while keeping user interactions local. However, transferring recommendation knowledge across services remains challenging due to disjoint item catalogs and uneven target-domain supervision. Frequently observed items provide abundant interaction evidence, whereas low-frequency items offer limited supervision, and client-specific preferences further introduce heterogeneity into semantic adaptation. Existing approaches commonly employ shared transformations or client-level personalization without explicitly accounting for differences in item-level supervision. To address these challenges, we propose FedPASA, a reliability-aware semantic adaptation framework for cross-domain federated recommendation. FedPASA introduces Popularity-Gated Dual-path Routing (PGDR), which combines two residual adaptation paths using item-frequency-derived weights. A private client profile further conditions the adapted representations through feature-wise scaling and shifting, while AutoAnchor incorporates pretrained semantic information through catalog-level anchoring. Extensive experiments on real-world cross-domain recommendation benchmarks demonstrate the effectiveness of FedPASA. The proposed framework achieves superior average cross-domain recommendation performance over competitive baselines under both sampled and full-catalog ranking. Further analyses demonstrate that frequency-aware routing improves recommendation accuracy beyond additional adapter capacity, while client conditioning and semantic anchoring provide complementary adaptation capabilities.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Distributed computing methodologies.*

Keywords

Federated Recommendation; Cross-Domain Recommendation; Semantic Transfer; Long-tail Recommendation

ACM Reference Format:

Anonymous Author(s). 2027. FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation. In *Proceedings of the ACM Web Conference 2027 (WWW '27).* ACM, New York, NY, USA, 12 pages.

1 Introduction

Recommender systems learn user–item preferences from interaction histories [9] and support online services. Centralized training, however, raises concerns about sharing these histories. Federated recommendation (FR) keeps interactions local and aggregates model parameters [1, 23]. Recent work develops client co-clustering [10], distributed heterogeneous-graph learning [33], and communication-efficient low-rank updates [25], advancing collaborative recommendation without pooling raw histories.

Cross-domain transfer is another longstanding need: platforms wish to reuse a mature catalog to bootstrap a new service with limited behavioral evidence. Many cross-domain methods transfer preferences through overlapping users or aligned representations [3, 22, 34]. Such links may be unavailable across services, while federated constraints prevent pooling user histories. We study disjoint item catalogs without requiring linked users between source and target domains.

Text-based transferable representations offer an alternative to identifier alignment [11, 13, 38]. However, semantic representations alone do not resolve uneven target supervision. Interaction imbalance and conformity can distort representation learning [34, 35]: frequent items supply many ranking constraints, whereas rare items provide little adaptation evidence. Clients also observe different semantic regions, so a globally rare item may be familiar to one client yet unseen by another. Excessive adaptation can move sparse items away from useful pretrained semantics. Adaptation should therefore account for item-level evidence, private client context, and the amount of semantic prior to preserve.

Hierarchical personalization, personalized adapters, and expert mixtures address client variation [18, 32, 36]. Client-level personalization alone does not specify how items with different training frequencies should share adaptation paths: a shared transformation uses the same parameterization for all items, while adding an adapter increases capacity without determining its allocation. We instead use item frequency as an observable proxy for target supervision, route items between two residual paths, and condition the routed representation on the local interaction profile.

To address the challenges mentioned above, we propose FedPASA, which combines three operations in a shared semantic adapter. Specifically, Popularity-Gated Dual-path Routing (PGDR) mixes a source-initialized path and a target-initialized path using a frequency-derived weight; client conditioning produces feature-wise scales and shifts from a locally pooled semantic profile; in addition, AutoAnchor interpolates the resulting representation with the frozen text embedding, with the anchor strength set by a catalog-level user-to-item ratio. Both residual paths are updated on the target domain, while source initialization provides a starting point rather than a fixed representation.

We evaluate this design on multiple real-world datasets, with controls that separate frequency assignment, model capacity, and final personalization. Our contributions are summarized as follows:

- We propose FedPASA, a reliability-aware semantic adaptation framework that adjusts item representations according to evidence strength at the item, client, and catalog levels. Our framework is motivated by an analysis of the failure of uniform adaptation in existing federated semantic

WWW '27, Dublin, Ireland
2027.

transfer methods under uneven target supervision: a single shared transformation underfits well-supported frequent items while overfitting sparsely observed items. The proposed framework applies to different backbone encoders.
- We design PGDR to combine two trainable residual adaptation paths using frequency-derived weights, without an additional learned routing network. Client conditioning adjusts the combined representations using local semantic profiles, while AutoAnchor controls their mixture with the frozen semantic prior through a catalog-level rule.
- We conduct extensive experiments on multiple real-world datasets. With the same text encoder, FedPASA achieves higher average cross-domain recommendation accuracy than the evaluated baselines. Comparisons that retain the same network structure and parameter count further show that frequency-derived mixing weights outperform using a common pair of weights for all items or shuffling the item–weight assignments. Ablation studies further characterize the effects of individual components.

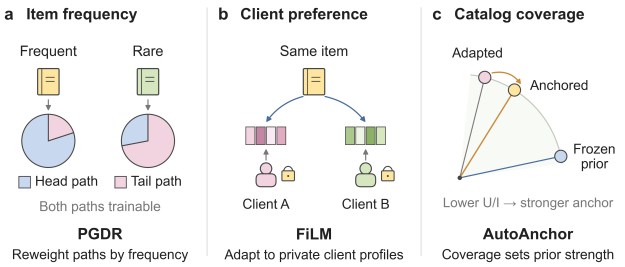


Figure 1: Three adaptation controls: (a) frequency-based path mixing; (b) private-profile conditioning; (c) catalog-dependent prior preservation.

2 Related Work

2.1 Federated Semantic Transfer

FR learns shared parameters from decentralized interactions [1, 23]; proximal optimization and local components address heterogeneity [2, 5, 14, 15]. Transferring item-ID models across disjoint catalogs requires alignment or content. TransFR combines pretrained language representations, adapter tuning, and local personalization [38]. FedDCSR disentangles shared and domain-specific sequential factors [39]; PFCR and FFMSR transfer text-derived content [6, 20]; FedCRF uses semantic graphs and contrastive fusion [7]. FeDecider personalizes LoRA aggregation with client-specific directional weights [8]. Beyond sequence modeling, fusion, and aggregation, FedPASA studies item-level supervision allocation within a shared semantic adapter.

2.2 Personalization and Preservation

FedPA combines private user adapters, shared group adapters, and adaptive gating [36]; FedVLR personalizes multimodal expert mixtures [18]. ESAM addresses long-tail imbalance [4], MeLU adapts user preferences [12], and multi-gate MoE learns input-dependent expert combinations [21]. PGDR instead derives deterministic routing weights from target item frequency. Learned, constant, and shuffled routing provide comparisons that isolate this allocation rule.

FiLM applies feature-wise scales and shifts [27]; FedPASA generates them from a private client profile within a shared adapter. L2-SP and DELTA preserve source information through parameter or feature penalties [16, 17]. AutoAnchor instead interpolates the normalized ranking representation with the frozen semantic prior. Its fixed catalog-level coverage heuristic is motivated by bias–variance and geometric analyses and assessed through ranking experiments.

3 FedPASA

3.1 Problem Setup and Overview

We consider cross-domain federated recommendation with disjoint item catalogs. Let $\mathcal{C}$, $\mathcal{U}$, and $\mathcal{I}$ denote the sets of clients, users, and target items, respectively. Each client c maintains a subset of users $\mathcal{U}_c$ and their interaction histories locally. A frozen text encoder maps each item i to a unit-normalized semantic representation $x_i \in \mathbb{R}^d$, providing a common representation space across domains. Each user u is associated with a private embedding $e_u \in \mathbb{R}^d$ that remains on its corresponding client.

FedPASA adapts the frozen semantic representations through three mechanisms: Popularity-Gated Dual-path Routing (PGDR), Private-Profile Client Conditioning, and Catalog-Level Semantic Anchoring (AutoAnchor). Source-domain training initializes one residual adaptation path, while the second path is randomly initialized on the target domain. Both paths remain trainable during target-domain federated optimization.

For user u on client c, the recommendation score is defined as

$$s(u,i) = e_u^\top z_{c,i}^{\text{anchor}}. \tag{1}$$

The model is optimized using Bayesian Personalized Ranking (BPR) [29]:

$$\mathcal{L}_{\text{BPR}} = -\mathbb{E}_{(u,i,j)} \log \sigma\big(\tau[s(u,i) - s(u,j)]\big),$$

where i is an observed positive item, j a sampled negative item, and τ controls the score scale.

Figure 2 illustrates the overall FedPASA framework, and Table 6 summarizes the notation. The forward computation consists of three consecutive stages:

$$x_i \xrightarrow{\text{routing}} h_i \xrightarrow[\text{and residual}]{\text{FiLM}} z_{c,i} \xrightarrow{\text{anchoring}} z_{c,i}^{\text{anchor}}. \tag{2}$$

3.2 Popularity-Gated Dual-path Routing

Target items exhibit substantially different interaction frequencies, resulting in different amounts of supervision for semantic adaptation. PGDR explicitly uses item frequency to determine the relative contributions of two residual adaptation paths, enabling frequency-dependent representation learning within a shared model.

Let $\text{freq}(i)$ denote the number of target-domain training interactions containing item i. We first construct an inverse-frequency score $\tilde{q}_i = 1/\sqrt{\text{freq}(i)+1}$ and normalize it over the target catalog:

$$q_i = \frac{\tilde{q}_i - \min_j \tilde{q}_j}{\max_j \tilde{q}_j - \min_j \tilde{q}_j + \epsilon}. \tag{3}$$

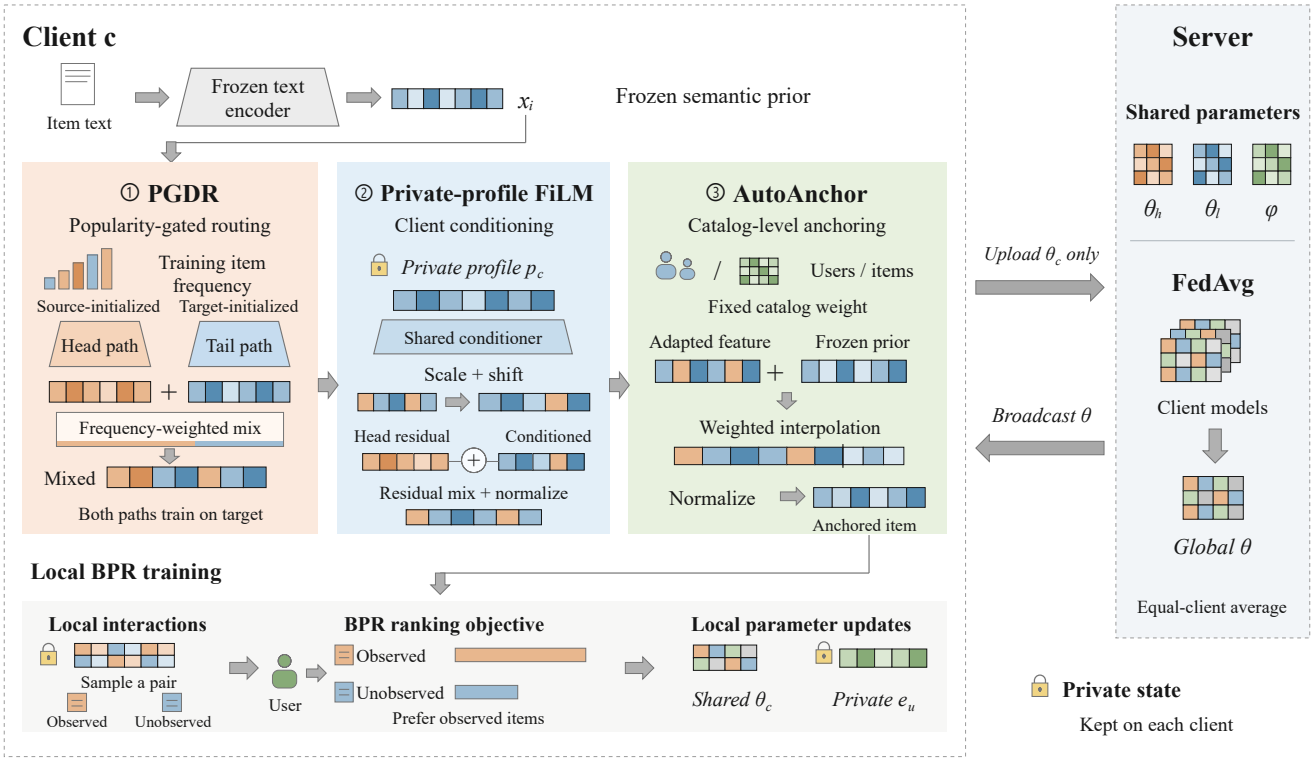


Figure 2: FedPASA framework. PGDR combines source-initialized and target-initialized residual adapters according to item frequency. Private-profile FiLM conditions the resulting representations on local interaction semantics, while AutoAnchor integrates the frozen semantic prior. Clients optimize shared model parameters and private user embeddings using BPR, and the server aggregates the shared parameters through FedAvg.

A threshold t and exponent κ further reshape the routing score:

$$q_i' = \mathrm{clip}\left(\frac{q_i - t}{1 - t}, 0, 1\right)^{\kappa}.$$

The resulting score q_i' increases as item frequency decreases and controls the relative contribution of the target-initialized path.

PGDR contains two residual adapters, denoted by A_h and A_l. The head adapter A_h is initialized from the source-domain adapter, providing transferred semantic adaptation knowledge. The tail adapter A_l is randomly initialized on the target domain and learns complementary transformations from target interactions. Each adapter contains two sequential residual blocks, $v \leftarrow v + F_\ell(v)$, where F_ℓ is a bottleneck multilayer perceptron, followed by ℓ_2 normalization. Both adapters are optimized during target-domain federated training.

The routed item representation is

$$h_i = (1 - q_i')\, A_h(x_i) + q_i'\, A_l(x_i). \tag{4}$$

Frequently observed items place greater weight on the source-initialized head path, whereas lower-frequency items receive a larger contribution from the target-initialized tail path. The mixed representation h_i is subsequently passed to the client-conditioning module.

3.3 Private-Profile Client Conditioning

Item frequency characterizes supervision at the item level, while clients may observe substantially different semantic regions of the target catalog. To incorporate this client-side heterogeneity, FedPASA constructs a private semantic profile from each client's local interaction history and uses it to condition the routed representation.

For client c, let $\mathcal{I}_c$ denote the multiset of items appearing in its training interactions. Repeated interactions retain their multiplicity in the profile construction. The client profile is defined as

$$p_c = \mathrm{norm}\left(\frac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} x_i\right). \tag{5}$$

A shared bottleneck network $g_\phi : \mathbb{R}^d \to \mathbb{R}^{2d}$ maps the profile into feature-wise FiLM parameters:

$$(\gamma_c, \beta_c) = g_\phi(p_c). \tag{6}$$

The routed representation h_i is then modulated as

$$\begin{aligned} m_{c,i} &= h_i \odot (1 + \lambda \tanh \gamma_c) + \lambda \beta_c, \\ z_{c,i} &= \mathrm{norm}\big((1 - \alpha) A_h(x_i) + \alpha m_{c,i}\big). \end{aligned} \tag{7}$$

The coefficient λ controls the strength of client-dependent modulation. The tanh transformation bounds the multiplicative factor to $[1-\lambda, 1+\lambda]$, while the additive term provides learned client-specific

feature shifts. The coefficient $\alpha \in [0, 1]$ controls the contribution of the conditioned representation relative to the head-path residual.

For the conditioning module, only the shared parameters ϕ participate in federated communication. The client profile p_c is computed from local training interactions and remains on the client.

3.4 Catalog-Level Semantic Anchoring

After frequency-aware routing and client conditioning, FedPASA further integrates the frozen semantic representation into the final ranking representation. AutoAnchor determines the contribution of this semantic prior according to target-catalog coverage.

We characterize target-catalog coverage using the user-to-item ratio

$$\eta_T = |\mathcal{U}_T|/|\mathcal{I}_T|,$$

where $|\mathcal{U}_T|$ denotes the number of observed target-domain users and $|\mathcal{I}_T|$ denotes the target catalog size. Given coverage thresholds $\tau_l < \tau_h$ and maximum anchor weight ρ, AutoAnchor computes

$$\begin{aligned} \rho_{\text{eff}} &= \rho \cdot \text{clip}\Big(\frac{\tau_h - \eta_T}{\tau_h - \tau_l}, 0, 1\Big), \\ z_{c,i}^{\text{anchor}} &= \text{norm}\big((1 - \rho_{\text{eff}})\, z_{c,i} + \rho_{\text{eff}}\, x_i\big). \end{aligned} \tag{8}$$

The effective weight ρ_{eff} controls the contribution of the frozen semantic prior. When $\eta_T \le \tau_l$, the maximum anchor weight ρ is used. As η_T increases between τ_l and τ_h, the prior contribution decreases linearly. For $\eta_T \ge \tau_h$, the final representation relies directly on the adapted representation. Its weight is determined from target-domain catalog statistics and remains fixed throughout federated training and evaluation.

The default catalog-level gate introduces no additional trainable parameters. We additionally evaluate a per-item learnable alternative,

$$\rho_i = \rho\, \sigma(a_i),$$

where $a \in \mathbb{R}^{|\mathcal{I}|}$ contains item-specific logits included in federated aggregation. The corresponding comparison is reported in the anchor analysis (§4.5.1).

3.5 Federated Training and Local Personalization

FedPASA jointly learns shared semantic adaptation parameters across clients while maintaining private user representations locally. The shared trainable parameters are $\theta = \{\theta_h, \theta_l, \phi\}$, where θ_h and θ_l denote the parameters of the two residual adapters and ϕ denotes the parameters of the client-conditioning network. Each client c additionally maintains a private user-embedding table $E_c = \{e_u : u \in \mathcal{U}_c\}$.

During communication round r, the server broadcasts the current shared parameters $\theta^{(r)}$ to all participating clients. Each client initializes its local shared-model copy from $\theta^{(r)}$, retains its private user embeddings from the preceding round, and performs K local optimization steps using the BPR objective. The local optimization updates both the shared parameters θ_c and private user embeddings E_c. After local training, each client uploads only θ_c to the server. Under the default full-participation setting, the server performs equal-client FedAvg:

$$\theta^{(r+1)} = \frac{1}{|C|} \sum_{c \in C} \theta_c.$$

Algorithm 1 FedPASA training and local personalization

Require: Frozen item features, training histories, clients C, rounds R, local steps K, personalization steps K_p, proximal coefficient μ_p

For cross-domain training, optimize a source-domain residual adapter by source BPR and equal-client FedAvg
Initialize θ_h from the source adapter; randomly initialize θ_l and ϕ
Compute target routing scores q_i from target training frequencies
Compute each client profile p_c from its training interactions
Compute and fix the catalog-level anchor weight ρ_{eff}
Initialize private user tables $E_c = \{e_u : u \in \mathcal{U}_c\}$ for all clients
$\theta^{(0)} \leftarrow \{\theta_h, \theta_l, \phi\}$
for $r = 0, \ldots, R-1$ **do**
  Broadcast $\theta^{(r)}$ to all participating clients
  for all $c \in C$ **in parallel do**
    $\theta_c \leftarrow \theta^{(r)}$; retain E_c from the preceding round
    Take K local steps on $\mathcal{L}_{\text{BPR}}$, updating (θ_c, E_c)
    Keep E_c local and upload θ_c
  end for
  $\theta^{(r+1)} \leftarrow |C|^{-1} \sum_{c \in C} \theta_c$
end for
Preserve $\theta^{(R)}$ as the FedPASA base model
for all $c \in C$ **do**
  $\theta_c \leftarrow \theta^{(R)}$; retain the final E_c
  $\ell_c \leftarrow \mathcal{L}_{\text{BPR}} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2$
  Take K_p local personalization steps, updating (θ_c, E_c)
end for
return Personalized (θ_c, E_c) stored on each client

The federated adaptation stage optimizes the BPR objective without an additional proximal penalty.

After R communication rounds, the resulting global parameters $\theta^{(R)}$ define the shared FedPASA model. We refer to this model before final local personalization as *FedPASA base*. For final personalization, each client initializes its local model from $\theta^{(R)}$, retains its learned private user embeddings, and optimizes

$$\mathcal{L}_{\text{pers},c} = \mathcal{L}_{\text{BPR},c} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2.$$

The proximal term encourages local adaptation around the final global solution [14, 15]. During this stage, both θ_c and E_c are updated, while the proximal penalty is applied only to the shared-model parameters θ_c. Each client performs K_p local personalization steps. The resulting personalized parameters remain local and are not uploaded to the server.

Algorithm 1 summarizes the complete training procedure. Concrete optimization settings are provided in the experimental setup (§4.2).

Initialization and local optimization. Source-domain training uses one residual adapter together with source-domain user embeddings. The resulting adapter parameters initialize the target-domain head path. The tail path is randomly initialized. The final linear layer of the conditioning network is zero-initialized so that client modulation starts from the shared representation.

Target item frequencies are computed exclusively from target training interactions. Each client profile is computed from its own training-item multiset, with repeated interactions retained. Routing statistics, client profiles, and the catalog-level anchor weight are fixed throughout federated optimization, personalization, and evaluation.

Private user embeddings persist across communication rounds. AdamW optimizer states are reset at the beginning of each local optimization phase. During local training, users are sampled uniformly within a client and positive items are sampled uniformly

from the corresponding user's training history. Negative items are sampled from items outside the user's observed training interactions. Dropout is enabled during model updates and disabled during evaluation.

The communication process exchanges only the shared adaptation and conditioning parameters. Raw interaction histories, client profiles, and private user embeddings remain on their respective clients.

3.6 Analysis of the Adaptation Controls

We analyze the main adaptation mechanisms from three perspectives: the relationship between target-estimation reliability and semantic-prior weighting, the geometry of normalized semantic anchoring, and the interaction between popularity routing and client conditioning.

3.6.1 Bias–Variance Tradeoff of Prior Weighting.

Proposition 1 (Bias–variance tradeoff). *Let z^* denote an ideal target representation and x a semantic prior. Consider an estimated target representation $\hat{z} = z^* + \varepsilon$, where $\mathbb{E}\varepsilon = 0$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma^2/r$ with $r, \sigma^2 > 0$. Here, r characterizes the reliability of the target estimate: larger r corresponds to lower estimation variance. Let $b^2 = \|x - z^*\|_2^2$ denote the squared bias of the semantic prior. For the interpolated estimate $\tilde{z}_\omega = (1-\omega)\hat{z} + \omega x$, we have*

$$\mathbb{E}\|\tilde{z}_\omega - z^*\|_2^2 = (1-\omega)^2\sigma^2/r + \omega^2 b^2, \qquad \omega^* = \frac{\sigma^2}{\sigma^2 + rb^2}. \quad (9)$$

When $b > 0$, ω^* decreases as r increases.

The optimal weight increases with estimation noise and decreases with prior bias: when target evidence is limited, more of a useful prior should be retained. AutoAnchor operationalizes this principle at the catalog level by using target-catalog coverage to determine the contribution of the frozen semantic prior.

3.6.2 Geometry of Normalized Semantic Anchoring.

Proposition 2 (Normalized anchor geometry). *For fixed unit vectors x and z, define $w_\omega = (1-\omega)z + \omega x$, $y_\omega = w_\omega/\|w_\omega\|_2$, and $a = x^\top z$. For $0 \le \omega < 1/2$,*

$$\|w_\omega\|_2 \ge 1 - 2\omega > 0, \qquad \frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{\|w_\omega\|_2^3} \ge 0. \quad (10)$$

Therefore, $1 - x^\top y_\omega$ is non-increasing with respect to the anchor weight ω.

This proposition characterizes the geometric effect of normalized semantic interpolation: for a fixed adapted representation, increasing the anchor weight moves the final representation toward the frozen semantic prior. The experimental analysis (§4.5.1) further examines the representation geometry produced after federated optimization.

3.6.3 Client Conditioning. The multiplicative part of the FiLM transformation in Eq. (7) is $1 + \lambda \tanh \gamma_c$. Since $\tanh \gamma_c \in [-1, 1]$, the resulting feature-wise multiplicative factors satisfy $1 - \lambda \le 1 + \lambda \tanh \gamma_c \le 1 + \lambda$. Thus, λ directly controls the range of client-dependent multiplicative modulation, while the learned additive term $\lambda\beta_c$ shifts the routed representation according to the client's semantic profile.

Table 1: Dataset statistics after filtering, including training, validation, and test interactions.

Dataset	# Users	# Items	# Interactions
Music	12,315	10,680	339,923
Books	262,698	170,389	10,848,388
Movies	28,226	19,252	1,056,608

3.6.4 Interaction of Routing and Client Conditioning. PGDR and client conditioning act on the same semantic representation but control different aspects of adaptation. To make the interaction explicit, consider the case in which FiLM modulation is disabled, i.e., $\lambda = 0$. Combining Eqs. (4) and (7) gives

$$z_{c,i} = \mathrm{norm}\big((1 - \alpha q_i')A_h(x_i) + \alpha q_i' A_l(x_i)\big). \quad (11)$$

The effective coefficient of the target-initialized path is therefore $\alpha q_i'$. The item-frequency-derived score q_i' determines how the two adaptation paths are allocated to each item, while α controls the overall strength of this allocation. Even for the largest routing score $q_i' = 1$, the head path retains a contribution of $1 - \alpha$.

When FiLM is enabled, client-dependent feature scales and shifts are applied to the frequency-routed representation before normalization and semantic anchoring. Consequently, item frequency controls path combination, whereas the private client profile controls feature-level adjustment.

4 Experiments

We evaluate FedPASA through four research questions:

- **RQ1:** How does FedPASA compare with federated recommendation and semantic-transfer baselines?
- **RQ2:** How much does each component contribute beyond additional adapter capacity?
- **RQ3:** How do item frequency, client exposure, and catalog coverage explain the gains?
- **RQ4:** Does FedPASA remain effective under client heterogeneity, restricted computation, and alternative encoders?

4.1 Datasets

We evaluate cross-domain recommendation on three Amazon datasets. The benchmark contains all six ordered transfers among Amazon Music (CDs & Vinyl), Books, and Movies (Movies & TV) [26]; their item catalogs are disjoint and no linked users are required. Table 1 summarizes the retained data. Users with fewer than three interactions are removed; the last interaction is used for testing, the preceding interaction for validation, and all earlier interactions for training. Appendix B gives the evaluation-user counts and preprocessing details.

4.2 Experimental Settings

4.2.1 Implementation Details. We encode items with frozen DistilBERT-base-uncased representations ($d = 768$) and use two-block residual adapters with bottleneck dimension 64 and dropout 0.1. All adapter methods are optimized with AdamW using a learning rate of 10^{-3} and weight decay of 10^{-4}. Each BPR batch contains

Table 2: Sampled-ranking comparison on the first three transfers. FedPASA includes final local personalization. Bold and underline mark the best and second-best results; run counts are given in Appendix C.

Transfer	Music→Books		Music→Movies		Books→Movies	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [8]	0.2011	0.0714	<u>0.4278</u>	<u>0.1823</u>	<u>0.3981</u>	<u>0.1705</u>
FFMSR [20]	0.0273	0.0116	0.1354	0.0824	0.1231	0.0720
FedDCSR [39]	0.0254	0.0218	0.0924	0.0724	0.0924	0.0724
FedMF [1]	0.2005	0.0715	0.2127	0.0792	0.2127	0.0792
FedAvg [23]	0.1949	0.0672	0.3440	0.1388	0.3440	0.1388
FedProx [15]	0.1998	0.0705	0.3381	0.1351	0.3381	0.1351
FedAdam [28]	0.1986	0.0690	0.3398	0.1359	0.3398	0.1359
TransFR [38]	<u>0.2058</u>	<u>0.0723</u>	0.3292	0.1283	0.3330	0.1307
L2-SP [16]	0.2032	0.0718	0.3244	0.1244	0.3141	0.1229
FedPer [2]	0.2031	0.0714	0.3410	0.1353	0.3492	0.1397
pFedMe [5]	0.2047	0.0719	0.3284	0.1285	0.3368	0.1332
FedMoE [24]	0.2039	0.0707	0.3424	0.1355	0.3566	0.1444
FedPASA	**0.2719**	**0.1159**	**0.4980**	**0.2254**	**0.4948**	**0.2284**

Table 3: Sampled-ranking comparison on the remaining transfers and the six-transfer mean. Metrics and formatting follow Table 2.

Transfer	Books→Music		Movies→Books		Movies→Music		Six-transfer mean	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [8]	<u>0.5355</u>	<u>0.2283</u>	0.2058	<u>0.0734</u>	<u>0.5380</u>	<u>0.2284</u>	<u>0.3844</u>	<u>0.1590</u>
FFMSR [20]	0.1585	0.0841	0.0214	0.0094	0.1701	0.0992	0.1060	0.0598
FedDCSR [39]	0.1090	0.0832	0.0254	0.0218	0.1090	0.0832	0.0756	0.0591
FedMF [1]	0.2212	0.0826	0.2005	0.0715	0.2212	0.0826	0.2115	0.0778
FedAvg [23]	0.5074	0.2129	0.1949	0.0672	0.5074	0.2129	0.3488	0.1396
FedProx [15]	0.5024	0.2093	0.1998	0.0705	0.5024	0.2093	0.3468	0.1383
FedAdam [28]	0.5073	0.2098	0.1986	0.0690	0.5073	0.2098	0.3485	0.1382
TransFR [38]	0.5046	0.2121	0.2001	0.0705	0.5048	0.2115	0.3462	0.1376
L2-SP [16]	0.4996	0.2083	0.2017	0.0710	0.5040	0.2122	0.3412	0.1351
FedPer [2]	0.5052	0.2150	0.2040	0.0721	0.5027	0.2099	0.3509	0.1406
pFedMe [5]	0.5054	0.2131	0.2026	0.0712	0.5064	0.2117	0.3474	0.1383
FedMoE [24]	0.5088	0.2143	<u>0.2064</u>	0.0731	0.5107	0.2141	0.3548	0.1420
FedPASA	**0.5408**	**0.2403**	**0.2824**	**0.1249**	**0.5462**	**0.2439**	**0.4390**	**0.1965**

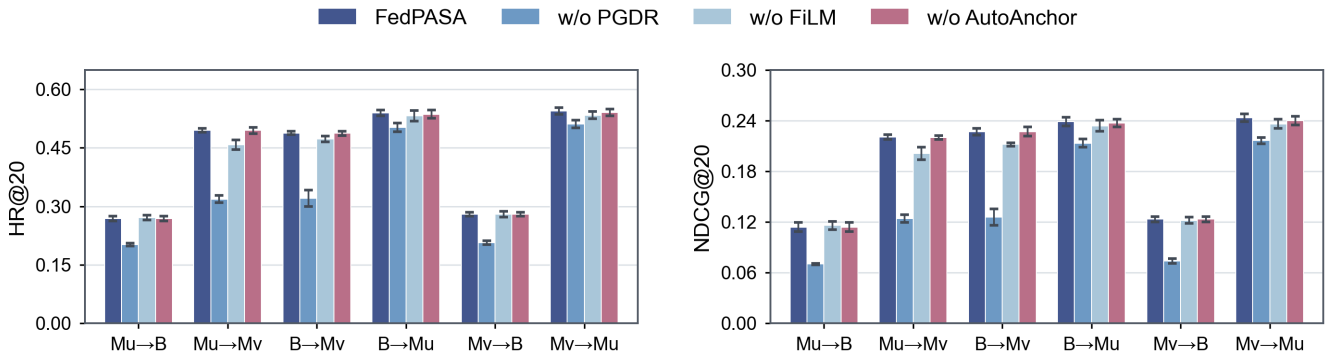


Figure 3: Component ablations under 99-negative ranking before personalization (five-seed means ± SD). Mu: Music; B: Books; Mv: Movies. PGDR contributes most; FiLM and AutoAnchor have target-dependent effects.

32 positives and three negatives per positive. Cross-domain experiments use five clients and train for 12×60 rounds/local steps on Music and Movies and 8×30 on Books. FedMF uses 64-dimensional

Table 4: Joint frequency–exposure ablation before local personalization under full-catalog ranking. C/U: low-frequency items locally seen/unseen; W: frequency > 5.

Model	C: low-frequency / seen		U: low-frequency / unseen		W: frequency > 5	
	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FedPASA base	**0.0160**	**0.0070**	**0.0347**	**0.0151**	**0.0102**	**0.0037**
w/o PGDR	0.0041	0.0010	0.0039	0.0013	0.0023	0.0007
w/o FiLM	0.0041	0.0018	0.0276	0.0123	0.0037	0.0014
w/o AutoAnchor	0.0082	0.0023	0.0144	0.0047	0.0039	0.0014

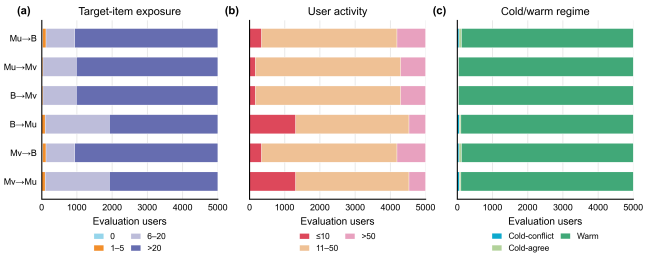


Figure 4: Evaluation-user composition: (a) target-item frequency, (b) user activity, and (c) cold/warm groups. Each transfer has 5,000 users. Mu/B/Mv: Music/Books/Movies.

factors and 25×80 rounds/local steps. FedPASA sets $\mu = 0$ during federation and performs $K_p = 40$ personalization steps with $\mu_p = .15$. Validation on Music⟶Movies selects $(\alpha, \kappa) = (.85, .5)$ and $(\lambda, \rho) = (.2, .3)$, which are fixed for all targets. We report HR and NDCG. Cross-domain evaluation uses a fixed 5,000-user subset with both 99-negative and full-catalog ranking; training and validation items are excluded from candidate sets. Common-backbone results are averaged over five runs. Additional baseline settings are given in Appendix C.

4.2.2 Baselines. We compare FedPASA with representative recent and classical federated recommendation baselines. **FeDecider** [8] uses LLM representations and client-specific directional weights to personalize LoRA aggregation. **TransFR** [38] transfers a source-initialized language-model adapter and performs local personalization. **FFMSR** [20] transfers text-derived semantic representations in federated recommendation. **FedDCSR** [39] disentangles shared and domain-specific factors for federated sequential recommendation. **FedMoE** [24] personalizes expert subsets through submodel search and modular aggregation. **FedAdam** [28] applies adaptive server-side optimization, while **FedProx** [15] adds a proximal term to constrain heterogeneous local updates. **pFedMe** [5] couples personalized client parameters to a shared model through a proximal objective. **FedPer** [2] combines a globally shared representation with client-specific personalization layers. **L2-SP** [16] regularizes target parameters toward their source initialization. **FedAvg** learns one shared target adapter by averaging client updates, and **FedMF** keeps user factors local while aggregating item factors on the server. In the common-backbone comparison, all methods use the same frozen DistilBERT item features [30], user partitions, and data splits; other baselines are evaluated with their corresponding datasets and ranking cutoffs.

4.3 Main Results and Discussion (RQ1)

Evidence-dependent adaptation improves transfer across disjoint catalogs. FedPASA leads both sampled-ranking metrics on all six transfers (Tables 2 and 3), with a mean HR improvement of 14.2% over FeDecider. The margins vary by target: gains are larger for Books and Movies, whereas the baselines are already competitive on Music. Frozen text features provide a shared semantic space, but do not determine how uneven target supervision should update that space. Frequency-dependent paths offer a plausible explanation for the additional gains; the ablations below isolate the contribution of each component.

The advantage extends beyond sampled candidates. FedPASA also leads the full-catalog comparison (Tables 7–9). The lower absolute accuracy is expected when many more items compete for the same top-ranked positions. Retaining the advantage supports improved discrimination beyond the sampled negatives, although Books remains difficult. This motivates examining both component effects and the item groups contributing to aggregate accuracy.

4.4 Ablation Study (RQ2)

Each variant retains BPR and the other modules and is evaluated before personalization (Figure 3; five-run variability). Tables 2–3 report personalized models.

(1) w/o PGDR. Removing the tail path makes routing independent of item frequency:

$$q'_i = 0, \qquad h_i = A_h(x_i). \tag{12}$$

Both metrics decrease on every transfer, consistent with losing item-dependent path allocation. PGDR also exceeds constant and shuffled routing (Table 16), which retain both paths but remove item dependence or break the frequency–weight correspondence. These controls link the gain to frequency assignment.

(2) w/o FiLM. Setting $\lambda = 0$ gives $m_{c,i} = h_i$ and

$$z_{c,i} = \mathrm{norm}\big((1 - \alpha q'_i)A_h(x_i) + \alpha q'_i A_l(x_i)\big). \tag{13}$$

Each item now shares one representation across clients. Lower accuracy on Movies and Music supports private-profile conditioning; near-tied Books results indicate a target-dependent benefit.

(3) w/o AutoAnchor. Removing the prior mixture yields

$$\rho_{\mathrm{eff}} = 0, \qquad z^{\mathrm{anchor}}_{c,i} = z_{c,i}. \tag{14}$$

The model relies entirely on adapted features. Books results remain identical because its catalog gate already disables anchoring; Music and Movies change modestly. Preservation is evaluated separately through representation drift.

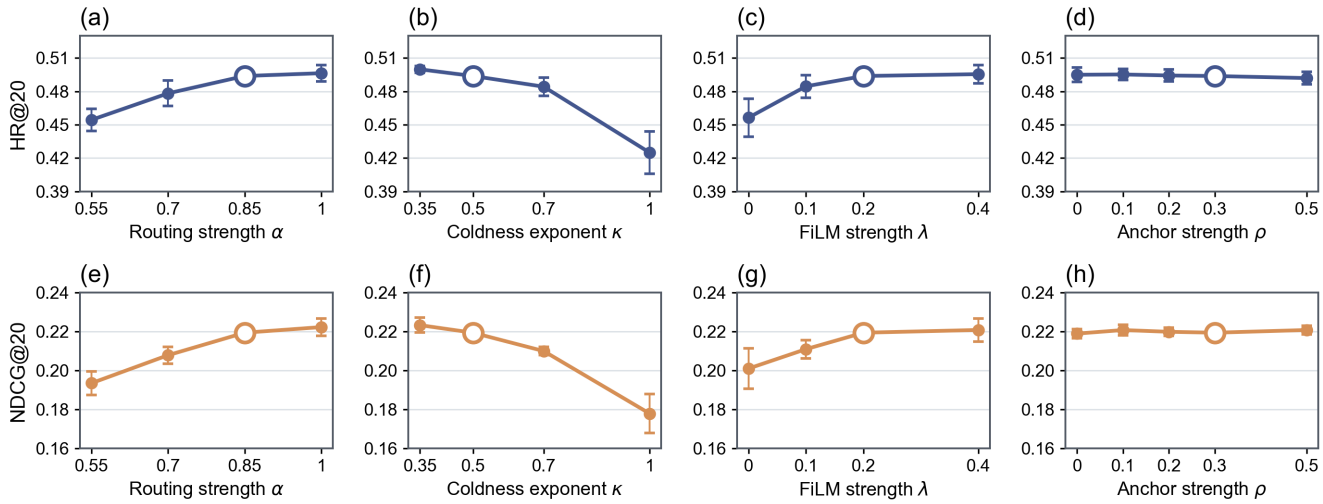


Figure 5: Parameter sensitivity on Music→Movies before personalization (99 negatives; three-seed means ± SD). Top: HR@20; bottom: NDCG@20. Each sweep varies one parameter. Open circles mark validation-selected defaults; vertical scales are shared within each row.

4.5 Evidence Mechanism Analysis (RQ3)

Table 4 compares the component ablations across item groups defined jointly by frequency and local exposure. Each removal lowers both ranking metrics relative to FedPASA base, including for locally seen and locally unseen low-frequency items.

Group improvements and aggregate gains are distinct. Figure 4 shows that frequent items and warm cases dominate evaluation, while user activity and exposure vary across target catalogs. Consequently, aggregate metrics weight the groups unequally: strong relative gains in a small low-frequency group need not dominate overall accuracy. The finer slices in Table 20 show that higher-frequency items contribute most of the full-ranking gain over FedAvg. Similar source-free and source-initialized averages (Table 16) further point to target-side adaptation as the main explanation.

4.5.1 AutoAnchor and representation drift. Preservation is measured separately from ranking through cosine distance:

$$D = \frac{1}{|C||\mathcal{I}|} \sum_{c \in C} \sum_{i \in \mathcal{I}} \left(1 - \cos(z_{c,i}^{\text{anchor}}, x_i)\right). \tag{15}$$

Anchoring slightly reduces cold-item distance but increases the all-item average after training (Table 26). Although Eq. (8) moves a fixed representation toward its prior before normalization, training changes the representation itself. The observed drift and ranking effects therefore depend on the item group and optimization outcome.

4.6 Generalization and Robustness (RQ4)

FedPASA remains effective under heterogeneous conditions. FedPASA exceeds TransFR in both metrics under all three non-IID Music→Movies partitions (Table 24). More concentrated activity makes prediction harder, but the advantage persists. The table also examines client counts, fixed computation, partial participation, and update noise.

The gains are not tied to one feature encoder. The results extend to all evaluated directions with LLaMA features (Table 24), supporting semantic adaptation beyond a particular encoder.

4.6.1 Sensitivity and statistical analysis. Figure 5 shows that increasing α improves both metrics, whereas increasing κ reduces them over the tested ranges. FiLM gains largely saturate near $\lambda = 0.2$. The flatter ρ response is consistent with weaker catalog-gated anchoring. Defaults are validation-selected; Table 23 lists the numerical results.

The positive confidence interval for paired HR improvements over the learned-gate baseline supports a repeatable aggregate advantage; Tables 10–13 report variability.

5 Conclusion

To address uneven target supervision in cross-domain federated recommendation, we propose FedPASA, a reliability-aware framework for adapting shared semantic representations across disjoint catalogs. PGDR allocates adaptation between two trainable residual paths according to item frequency, private-profile conditioning incorporates client-specific preferences, and AutoAnchor controls the contribution of the frozen semantic prior through catalog coverage. These operations jointly account for item evidence, local context, and prior preservation while keeping interaction histories local. Experiments across six transfer directions show improved average accuracy under both sampled and full-catalog ranking. Equal-capacity controls identify frequency-dependent routing as the main source of improvement, while component and group analyses characterize the contributions of conditioning and anchoring. The effects of these latter controls vary across item groups, suggesting that evidence-aware adaptation should accommodate local differences rather than assume uniform gains. Future work will explore finer-grained reliability estimates and adaptive conditioning to better handle conflicts between global frequency and local preferences.

References

[1] Muhammad Ammad-Ud-Din, Elena Ivannikova, Suleiman A Khan, Quang Oyomno, Qiang Fu, Kuan Ee Tan, and Adrian Flanagan. 2019. Federated Collaborative Filtering for Privacy-Preserving Personalized Recommendation System. *arXiv preprint arXiv:1901.09888* (2019).

[2] Manoj Ghuhan Arivazhagan, Vinay Aggarwal, Aaditya Kumar Singh, and Sunav Choudhary. 2019. Federated Learning with Personalization Layers. *arXiv preprint arXiv:1912.00818* (2019).

[3] Jiangxia Cao, Xixun Lin, Xin Cong, Jing Ya, Tingwen Liu, and Bin Wang. 2022. DisenCDR: Learning Disentangled Representations for Cross-Domain Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 267–277.

[4] Zhihong Chen, Rong Xiao, Chenliang Li, Gang Ye, Hao Sun, and Hongbo Deng. 2020. ESAM: Discriminative Domain Adaptation with Non-Displayed Items to Improve Long-Tail Performance. In *Proceedings of SIGIR*.

[5] Canh T. Dinh, Nguyen H. Tran, and Tuan Dung Nguyen. 2020. Personalized Federated Learning with Moreau Envelopes. In *Advances in Neural Information Processing Systems*, Vol. 33. 21394–21405.

[6] Lei Guo, Ziang Lu, Junliang Yu, Nguyen Quoc Viet Hung, and Hongzhi Yin. 2024. Prompt-Enhanced Federated Content Representation Learning for Cross-Domain Recommendation. *arXiv preprint arXiv:2401.14678* (2024).

[7] Lei Guo, Ting Yang, Hui Liu, Xu Yu, Xiaohui Han, and Xinhua Wang. 2026. FedCRF: A Federated Cross-Domain Recommendation Method with Semantic-Driven Deep Knowledge Fusion. *Information Processing & Management* (2026). arXiv:2604.17681 doi:10.1016/j.ipm.2026.104827

[8] Xinrui He, Ting-Wei Li, Tianxin Wei, Xuying Ning, Xinyu He, Wenxuan Bao, Hanghang Tong, and Jingrui He. 2026. FeDecider: An LLM-Based Framework for Federated Cross-Domain Recommendation. In *Proceedings of the ACM Web Conference*. doi:10.1145/3774904.3792294

[9] Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural Collaborative Filtering. In *Proceedings of the 26th International Conference on World Wide Web*. 173–182. doi:10.1145/3038912.3052569

[10] Xinrui He, Shuo Liu, Jacky Keung, and Jingrui He. 2024. Co-clustering for Federated Recommender System. In *Proceedings of the ACM Web Conference 2024*. 3821–3832. doi:10.1145/3589334.3645626

[11] Yupeng Hou, Zhankui He, Julian McAuley, and Wayne Xin Zhao. 2023. Learning Vector-Quantized Item Representation for Transferable Sequential Recommenders. In *Proceedings of the ACM Web Conference 2023*. 1162–1171. doi:10.1145/3543507.3583434

[12] Hoyeop Lee, Jinbae Im, Seongwon Jang, Hyunsouk Cho, and Sehee Chung. 2019. MeLU: Meta-Learned User Preference Estimator for Cold-Start Recommendation. In *Proceedings of KDD*.

[13] Jiacheng Li, Ming Wang, Jin Li, Jinmiao Fu, Xin Shen, Jingbo Shang, and Julian McAuley. 2023. Text Is All You Need: Learning Language Representations for Sequential Recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 1258–1267. doi:10.1145/3580305.3599519

[14] Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. 2021. Ditto: Fair and Robust Federated Learning Through Personalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. 6357–6368.

[15] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020. Federated Optimization in Heterogeneous Networks. *Proceedings of Machine Learning and Systems* 2 (2020), 429–450.

[16] Xuhong Li, Yves Grandvalet, and Franck Davoine. 2018. Explicit Inductive Bias for Transfer Learning with Convolutional Networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*. 2825–2834.

[17] Xingjian Li, Haoyi Xiong, Haozhe Wang, Yifan Rao, Liping Liu, Jie Chen, and Jun Huan. 2019. DELTA: Deep Learning Transfer using Feature Map with Attention for Convolutional Networks. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

[18] Zhiwei Li, Guodong Long, Jing Jiang, Chengqi Zhang, and Qiang Yang. 2026. Federated Vision-Language-Recommendation with Personalized Fusion. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 23337–23345. doi:10.1609/aaai.v40i28.39503

[19] Shuxia Liu, Shuang Xu, Wei Yu, Zhiqian Fu, Yongfeng Zhang, and Amelie Marian. 2021. FedCT: Federated Collaborative Transfer for Recommendation. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 716–725.

[20] Ziang Lu, Lei Guo, Xu Yu, Zhiyong Cheng, Xiaohui Han, and Lei Zhu. 2025. Federated Semantic Learning for Privacy-Preserving Cross-Domain Recommendation. *ACM Transactions on Information Systems* (2025). arXiv:2503.23026

[21] Jiaqi Ma, Zhe Zhao, Xinyang Yi, Jilin Chen, Lichan Hong, and Ed H. Chi. 2018. Modeling Task Relationships in Multi-Task Learning with Multi-Gate Mixture-of-Experts. In *Proceedings of KDD*.

[22] Tong Man, Huawei Shen, Xiaolong Jin, and Xueqi Cheng. 2017. Cross-Domain Recommendation: An Embedding and Mapping Approach. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*. 2464–2470.

[23] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 1273–1282.

[24] Hanzi Mei, Dongqi Cai, Ao Zhou, Shangguang Wang, and Mengwei Xu. 2024. FedMoE: Personalized Federated Learning via Heterogeneous Mixture of Experts. *arXiv preprint arXiv:2408.11304* (2024). doi:10.48550/arXiv.2408.11304

[25] Ngoc-Hieu Nguyen, Tuan-Anh Nguyen, Tuan Nguyen, Vu Tien Hoang, Dung D. Le, and Kok-Seng Wong. 2024. Towards Efficient Communication and Secure Federated Recommendation System via Low-rank Training. In *Proceedings of the ACM Web Conference 2024*. 3940–3951. doi:10.1145/3589334.3645702

[26] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying Recommendations Using Distantly-Labeled Reviews and Fine-Grained Aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 188–197.

[27] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. 2018. FiLM: Visual Reasoning with a General Conditioning Layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.

[28] Sashank J Reddi, Zachary Charles, Manzil Zaheer, Zachary Garrett, Keith Rush, Jakub Konecný, Sanjiv Kumar, and H Brendan McMahan. 2021. Adaptive Federated Optimization. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.

[29] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2009. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence (UAI)*. 452–461.

[30] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. *arXiv preprint arXiv:1910.01108* (2019).

[31] Ajit Paul Singh and Geoffrey J Gordon. 2008. Relational Learning via Collective Matrix Factorization. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 650–658.

[32] Jinze Wu, Qi Liu, Zhenya Huang, Yuting Ning, Hao Wang, Enhong Chen, Jinfeng Yi, and Bowen Zhou. 2021. Hierarchical Personalized Federated Learning for User Modeling. In *Proceedings of the Web Conference 2021*. 957–968. doi:10.1145/3442381.3449926

[33] Bo Yan, Yang Cao, Haoyu Wang, Wenchuan Yang, Junping Du, and Chuan Shi. 2024. Federated Heterogeneous Graph Neural Network for Privacy-preserving Recommendation. In *Proceedings of the ACM Web Conference 2024*. 3919–3929. doi:10.1145/3589334.3645693

[34] Wenhao Yang, Yingchun Jian, Yibo Wang, Shiyin Lu, Lei Shen, Bing Wang, Haihong Tang, and Lijun Zhang. 2024. Not All Embeddings are Created Equal: Towards Robust Cross-domain Recommendation via Contrastive Learning. In *Proceedings of the ACM Web Conference 2024*. 3195–3206. doi:10.1145/3589334.3645357

[35] Yuhao Yang, Chao Huang, Lianghao Xia, Chunzhen Huang, Da Luo, and Kangyi Lin. 2023. Debiased Contrastive Learning for Sequential Recommendation. In *Proceedings of the ACM Web Conference 2023*. 1063–1073. doi:10.1145/3543507.3583361

[36] Chunxu Zhang, Guodong Long, Hongkuan Guo, Xiao Fang, Yang Song, Zhaojie Liu, Guorui Zhou, Zijian Zhang, Yang Liu, and Bo Yang. 2024. Federated Adaptation for Foundation Model-Based Recommendations. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)*. 5453–5461.

[37] Chao Zhang, Guodong Long, Tianyi Zhou, Zijian Zhang, Peng Yan, and Baodi Yang. 2024. When Federated Recommendation Meets Cold-Start Problem: Separating Item Attributes and User Interactions. In *Proceedings of the ACM on Web Conference (WWW)*. 3632–3642.

[38] Honglei Zhang, Zhiwei Li, Haoxuan Li, Xin Zhou, Jie Zhang, and Yidong Li. 2026. TransFR: Transferable Federated Recommendation with Adapter Tuning on Pretrained Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 28212–28220.

[39] Huili Zhang, Dongxia Zheng, Xiang Yang, Juan Feng, and Qimei Liao. 2024. FedDCSR: Federated Cross-Domain Sequential Recommendation via Disentangled Representation Learning. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*. 535–543.

[40] Jingtao Zhao, Wenjie Wang, Cong Xu, See-Kiong Ng, and Tat-Seng Chua. 2025. A Federated Framework for LLM-based Recommendation. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 2852–2865.

A Proofs and Mechanism Analysis

We use the definitions and assumptions in §3.6.

A.1 Proof of Proposition 1

Let $v = \sigma^2/r > 0$. Expanding the error and using $\mathbb{E}\varepsilon = 0$ eliminates the cross term:

$$R(\omega) = (1-\omega)^2 v + \omega^2 b^2. \tag{16}$$

Thus $R'(\omega) = 2[(v+b^2)\omega - v]$ and $R''(\omega) = 2(v+b^2) > 0$. The unique minimizer on $[0,1]$ is $\omega^* = v/(v+b^2)$, including $\omega^* = 1$ when $b = 0$. Moreover,

$$\frac{d\omega^*}{dr} = -\frac{\sigma^2 b^2}{(\sigma^2 + rb^2)^2} \le 0. \tag{17}$$

The inequality is strict for $b > 0$; the estimator is unnormalized.

A.2 Proof of Proposition 2

Let $d_\omega = \|w_\omega\|$ and $n_\omega = (1-\omega)a + \omega$. The reverse triangle inequality gives $d_\omega \ge 1 - 2\omega > 0$, and

$$d_\omega^2 = (1-\omega)^2 + \omega^2 + 2\omega(1-\omega)a.$$

Using $n'_\omega = 1 - a$ and $d'_\omega = (2\omega - 1)(1-a)/d_\omega$, the quotient rule gives

$$\frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{d_\omega^3} \ge 0. \tag{18}$$

Here $|a| \le 1$ and x, z are fixed; different training trajectories are not ordered by this result.

A.3 Routing Gradients at Fixed Capacity

With FiLM disabled, Eq. (11) gives $v_i = (1-a_i)A_h(x_i) + a_i A_l(x_i)$, where $a_i = \alpha q'_i$ is fixed during training. For $g_i = \nabla_{v_i}\ell$, the chain rule gives

$$\begin{aligned} \nabla_{\theta_h}\ell &= (1-a_i) J_h(x_i)^\top g_i, \\ \nabla_{\theta_l}\ell &= a_i J_l(x_i)^\top g_i, \end{aligned} \tag{19}$$

where J_h, J_l are parameter Jacobians. Constant routing removes item-dependent gradient allocation; shuffling preserves capacity and the gate distribution but changes their association with item evidence.

A.4 Bounds for Client Conditioning

Write $s_c = \tanh\gamma_c$ and $m_c(h) = h + \lambda(s_c \odot h + \beta_c)$. Because h mixes unit vectors, $\|h\|_2 \le 1$. Using $\|s_c\|_\infty \le 1$ and $\lambda \ge 0$ gives

$$\begin{aligned} \|m_c(h) - h\|_2 &\le \lambda\big(\|s_c \odot h\|_2 + \|\beta_c\|_2\big) \\ &\le \lambda(1 + \|\beta_c\|_2). \end{aligned} \tag{20}$$

For fixed client parameters, the additive shift cancels:

$$\|m_c(h) - m_c(\tilde{h})\|_2 \le (1+\lambda)\|h - \tilde{h}\|_2. \tag{21}$$

Both bounds concern pre-normalization modulation, not ranking accuracy.

B Dataset Statistics

Table 5 reports retained users after the three-interaction filter; their last two events form validation and test sets. AutoAnchor divides this user count by catalog size.

Table 5: Retained datasets. N: all interactions; 5,000 users per target are evaluated.

Dataset	Users	Items	N
Music	12,315	10,680	339,923
Books	262,698	170,389	10,848,388
Movies	28,226	19,252	1,056,608

Ranking masks training and validation items (§4.2). DistilBERT uses normalized masked-mean features with at most 96 tokens; LLaMA-2-7B features use a fixed signed block projection to 512 dimensions.

Metric aggregation. For held-out rank r_u, HR@K averages $\mathbf{1}[r_u \le K]$ and NDCG@K averages $\mathbf{1}[r_u \le K]/\log_2(r_u+1)$. For a partition into nonempty groups $\mathcal{G}$,

$$M = \sum_{G \in \mathcal{G}} \frac{|G|}{|\mathcal{U}_{\text{eval}}|} M_G, \qquad M \in \{\text{HR@}K, \text{NDCG@}K\}.$$

Thus group size weights each group mean M_G in the aggregate.

Table 6: Notation used in FedPASA.

Symbol	Definition
x_i, e_u	Frozen unit item vector; private user vector
A_h, A_l, F_ℓ	Head/tail adapters; residual bottleneck MLP
q_i, q'_i, t, κ	Coldness scores, threshold, and exponent
$h_i, m_{c,i}, z_{c,i}$	Routed, modulated, and normalized features
p_c, g_ϕ	Private profile and shared conditioner
α, λ	Mixture and client-modulation strengths
$\eta_T, \rho, \rho_{\text{eff}}, \tau_l, \tau_h$	Coverage, anchor weights, and thresholds
θ, μ, μ_p	Shared parameters and proximal coefficients
r, b^2, σ^2	Reliability, prior bias, and noise scale

C Full Ranking and Run-to-Run Variation

Tables 7–9 report full ranking: H/N denote HR@20/NDCG@20; bold/underline mark best/second-best results. FeDecider uses three runs; others use five. FedPASA includes personalization.

Table 7: Full ranking on M→B, M→V, B→V.

	M→B		M→V		B→V	
Method	H	N	H	N	H	N
FedMF [1]	<u>0.0003</u>	<u>0.0001</u>	0.0022	0.0009	0.0022	0.0009
FedAvg [23]	0.0002	0.0000	0.0037	0.0015	0.0037	0.0015
FedProx [15]	0.0001	0.0000	0.0041	0.0015	0.0041	0.0015
FedAdam [28]	0.0001	0.0000	0.0033	0.0011	0.0033	0.0011
TransFR [38]	0.0001	0.0000	0.0037	0.0012	0.0030	0.0011
L2-SP [16]	0.0000	0.0000	0.0029	0.0011	0.0028	0.0011
FedPer [2]	0.0001	0.0000	0.0034	0.0012	0.0040	0.0014
pFedMe [5]	0.0001	0.0000	0.0029	0.0009	0.0034	0.0012
FedMoE [24]	0.0001	0.0000	0.0041	0.0015	0.0045	0.0016
FeDecider [8]	0.0001	0.0000	<u>0.0071</u>	<u>0.0025</u>	<u>0.0055</u>	<u>0.0019</u>
FedPASA	**0.0017**	**0.0006**	**0.0126**	**0.0046**	**0.0140**	**0.0050**

Table 8: Full ranking on B→M, V→B, V→M.

	B→M		V→B		V→M	
Method	H	N	H	N	H	N
FedMF [1]	0.0024	0.0009	<u>0.0003</u>	<u>0.0001</u>	0.0024	0.0009
FedAvg [23]	0.0125	0.0043	0.0002	0.0000	0.0125	0.0043
FedProx [15]	0.0130	0.0047	0.0001	0.0000	0.0130	0.0047
FedAdam [28]	0.0110	0.0041	0.0001	0.0000	0.0110	0.0041
TransFR [38]	0.0126	0.0047	0.0000	0.0000	0.0132	0.0050
L2-SP [16]	0.0132	0.0046	0.0001	0.0000	0.0132	0.0049
FedPer [2]	0.0127	0.0045	0.0002	0.0001	0.0119	0.0042
pFedMe [5]	0.0125	0.0046	0.0000	0.0000	0.0132	0.0049
FedMoE [24]	0.0134	<u>0.0050</u>	0.0002	0.0001	0.0137	0.0052
FeDecider [8]	<u>0.0136</u>	0.0049	0.0001	0.0001	<u>0.0145</u>	<u>0.0055</u>
FedPASA	**0.0177**	**0.0065**	**0.0018**	**0.0009**	**0.0188**	**0.0069**

Table 9: Full-ranking means over six transfers.

Method	HR@20	NDCG@20
FedMF [1]	0.0016	0.0006
FedAvg [23]	0.0055	0.0019
FedProx [15]	0.0057	0.0021
FedAdam [28]	0.0048	0.0018
TransFR [38]	0.0054	0.0020
L2-SP [16]	0.0054	0.0019
FedPer [2]	0.0054	0.0019
pFedMe [5]	0.0053	0.0019
FedMoE [24]	0.0060	0.0022
FeDecider [8]	0.0068	0.0025
FedPASA	**0.0111**	**0.0041**

Sampled-ranking variability is reported below as mean ± sample SD. For each method, the SD of the six-transfer mean is computed across run-level means.

Table 10: Sampled HR@20 variability: baselines.

Transfer	TransFR	FedMoE	FeDecider
M→B	0.2058 ± 0.0051	0.2039 ± 0.0032	0.2011 ± 0.0023
M→V	0.3292 ± 0.0065	0.3424 ± 0.0120	0.4278 ± 0.0055
B→V	0.3330 ± 0.0210	0.3566 ± 0.0077	0.3981 ± 0.0131
B→M	0.5046 ± 0.0127	0.5088 ± 0.0056	0.5355 ± 0.0240
V→B	0.2001 ± 0.0071	0.2064 ± 0.0059	0.2058 ± 0.0029
V→M	0.5048 ± 0.0074	0.5107 ± 0.0085	0.5380 ± 0.0214
Mean	0.3462 ± 0.0066	0.3548 ± 0.0042	0.3844 ± 0.0070

Table 11: Sampled HR@20 variability: FedPASA variants.

Transfer	FedPASA base	FedPASA
M→B	0.2692 ± 0.0063	0.2719 ± 0.0068
M→V	0.4950 ± 0.0055	0.4980 ± 0.0085
B→V	0.4881 ± 0.0039	0.4948 ± 0.0053
B→M	0.5394 ± 0.0072	0.5408 ± 0.0126
V→B	0.2796 ± 0.0053	0.2824 ± 0.0052
V→M	0.5449 ± 0.0085	0.5462 ± 0.0094
Mean	0.4360 ± 0.0044	0.4390 ± 0.0058

Table 12: Sampled NDCG@20 variability: baselines.

Transfer	TransFR	FedMoE	FeDecider
M→B	0.0723 ± 0.0032	0.0707 ± 0.0010	0.0714 ± 0.0014
M→V	0.1283 ± 0.0028	0.1355 ± 0.0035	0.1823 ± 0.0055
B→V	0.1307 ± 0.0090	0.1444 ± 0.0027	0.1705 ± 0.0093
B→M	0.2121 ± 0.0044	0.2143 ± 0.0031	0.2283 ± 0.0099
V→B	0.0705 ± 0.0030	0.0731 ± 0.0024	0.0734 ± 0.0027
V→M	0.2115 ± 0.0042	0.2141 ± 0.0059	0.2284 ± 0.0095
Mean	0.1376 ± 0.0027	0.1420 ± 0.0019	0.1590 ± 0.0042

Table 13: Sampled NDCG@20 variability: FedPASA variants.

Transfer	FedPASA base	FedPASA
M→B	0.1140 ± 0.0053	0.1159 ± 0.0054
M→V	0.2204 ± 0.0031	0.2254 ± 0.0053
B→V	0.2268 ± 0.0042	0.2284 ± 0.0032
B→M	0.2389 ± 0.0051	0.2403 ± 0.0076
V→B	0.1232 ± 0.0034	0.1249 ± 0.0035
V→M	0.2432 ± 0.0046	0.2439 ± 0.0069
Mean	0.1944 ± 0.0032	0.1965 ± 0.0035

Baseline settings. The FedMoE comparison uses two shared adapters with a learned gate, without client-specific submodel search or modular aggregation. FedDCSR reports target-domain results, so transfers to the same target share a value. FFMSR retains its corresponding evaluation setting.

D Additional Baseline Comparisons and Ablations

Table 14: Amazon Music/Books/Movies HR@10 [38].

Method	M→B	M→V	B→V	Mean
CMF [31]	0.1833	0.1378	0.1455	0.1555
EMCDR [22]	0.2149	0.1893	0.1652	0.1898
DisenCDR [3]	0.2243	0.1988	0.1912	0.2048
FCF [1]	0.1380	0.1154	0.1004	0.1179
IFedRec [37]	0.1981	0.1810	0.1703	0.1831
FELLRec [40]	0.2003	0.1844	0.1731	0.1859
FedCT [19]	0.2034	0.1765	0.1585	0.1795
FedDCSR [39]	0.2217	0.1881	0.1719	0.1939
TransFR [38]	**0.2335**	**0.2123**	**0.2089**	**0.2182**

Table 15: Amazon Music/Books/Movies NDCG@10 [38].

Method	M→B	M→V	B→V	Mean
CMF [31]	0.0955	0.0764	0.0662	0.0794
EMCDR [22]	0.1121	0.0901	0.0705	0.0909
DisenCDR [3]	0.1182	0.0923	0.0821	0.0975
FCF [1]	0.0663	0.0583	0.0421	0.0556
IFedRec [37]	0.0834	0.0673	0.0497	0.0668
FELLRec [40]	0.0995	0.0812	0.0831	0.0879
FedCT [19]	0.1122	0.0899	0.0845	0.0955
FedDCSR [39]	0.1085	0.0892	0.0732	0.0903
TransFR [38]	**0.1133**	**0.0934**	**0.0893**	**0.0987**

Table 16: Component and capacity controls: six-transfer means over five runs. Only the final row includes personalization.

Model	HR@20	NDCG@20
FedPASA base	0.4360	0.1944
w/o PGDR	0.3434	0.1373
w/o FiLM	0.4245	0.1867
w/o AutoAnchor	0.4346	0.1934
Constant routing	0.3481	0.1404
Random routing	0.3381	0.1350
Per-item anchor	0.4365	0.1950
Source-free base	0.4362	0.1958
FedMoE [24]	0.3548	0.1420
Personalized	0.4390	0.1965

Table 17: FFMSR and PFCR recommendation performance. R: recall; N: NDCG.

FFMSR [20]				
Domain	R10	R50	N10	N50
Office	0.1312	0.2069	0.0890	0.1054
Arts	0.1257	0.2377	0.0711	0.0955
OnlineRetail	0.1655	0.4110	0.0763	0.1300
Pantry	0.0695	0.1818	0.0326	0.0568
PFCR (full prompting) [6]				
Domain	R10	R50	N10	N50
Office	0.1206	0.1938	0.0782	0.0941
Arts	0.1153	0.2199	0.0677	0.0904
OnlineRetail	0.1561	0.3982	0.0739	0.1266
Pantry	0.0616	0.1591	0.0293	0.0503

Table 18: FedCRF full-ranking Recall [7].

Pair / target	Recall@10	Recall@20
Kitchen–Food: Kitchen	0.04465	0.06601
Kitchen–Food: Food	0.11078	0.16105
Care–Beauty: Care	0.07115	0.09897
Care–Beauty: Beauty	0.09342	0.13384
Food–OnlineRetail: Food	0.11142	0.16185
Food–OnlineRetail: OnlineRetail	0.22864	0.31488

Table 19: FedCRF full-ranking NDCG [7].

Pair / target	NDCG@10	NDCG@20
Kitchen–Food: Kitchen	0.02523	0.03112
Kitchen–Food: Food	0.06708	0.08193
Care–Beauty: Care	0.04250	0.05001
Care–Beauty: Beauty	0.05398	0.06513
Food–OnlineRetail: Food	0.06713	0.08189
Food–OnlineRetail: OnlineRetail	0.19232	0.23368

Table 20: Fine-grained frequency slices (99-negative ranking; unpersonalized models).

Frequency	Method	HR@20	NDCG@20
Unseen	TransFR [38]	0.3020	0.1070
	FedMoE [24]	0.2460	0.0927
	FedPASA base	0.5360	0.3783
1–5	TransFR [38]	0.3711	0.1562
	FedMoE [24]	0.3915	0.1662
	FedPASA base	0.3439	0.2045
6–20	TransFR [38]	0.3465	0.1395
	FedMoE [24]	0.3541	0.1438
	FedPASA base	0.2631	0.0990
>20	TransFR [38]	0.3455	0.1361
	FedMoE [24]	0.3544	0.1404
	FedPASA base	0.4956	0.2258

The unseen slice contains 20 nonempty evaluations (mean support: 13 users); other slices contain 30. Evaluations with zero hits are included.

E Hyperparameters and Robustness Analysis

Table 21: Validation-only selection on Music→Movies (mean sampled HR@20 over three runs; eight rounds, 30 local steps). Stage 1 fixes $\lambda = 0.4, \rho = 0.2$; Stage 2 fixes the selected $\alpha = 0.85, \kappa = 0.5$. The final choice is $(0.85, 0.5, 0.2, 0.3)$.

$\alpha\backslash\kappa$	0.5	0.7	1.0
0.4	0.2807	0.2640	0.2517
0.55	0.3261	0.2871	0.2626
0.7	0.3657	0.3139	0.2721
0.85	0.3851	0.3541	0.2883

$\lambda\backslash\rho$	0.1	0.2	0.3
0.2	0.3917	0.3917	0.3925
0.4	0.3879	0.3851	0.3893
0.6	0.3842	0.3874	0.3817

E.1 Configuration

Table 22: Default configuration. Sensitivity experiments vary one setting at a time.

Setting	Value
Frozen encoder	DistilBERT-base-uncased, $d = 768$
Adapter	2 residual blocks; bottleneck 64; dropout 0.1
Optimizer	AdamW; lr 10^{-3}; weight decay 10^{-4}
BPR batches	32 positives; 3 negatives/positive; logit scale 10
Federation	Cross: 12×60 rounds/steps; target Books: 8×30
Clients	5
Mixing / FiLM	$\alpha = 0.85$, $t = 0$, $\kappa = 0.5$, $\lambda = 0.2$
Anchor	$\rho = 0.3$; $[\tau_l, \tau_h] = [1, 1.5]$
Proximal terms	FedProx: 0.1; L2-SP: 0.01; base FedPASA: 0
Final personalization	40 local steps, $\mu_p = 0.15$ (full FedPASA and pFedMe)
FedMF baseline	64-dimensional factors; 25 rounds, 80 steps/round

Table 23: HR@20/NDCG@20 for Figure 5 (Music→Movies, three runs). (d): selected default.

Factor	Value	HR@20	NDCG@20
α	0.55	0.4543	0.1935
	0.7	0.4782	0.2079
	0.85	0.4938	0.2194 (d)
	1.0	0.4963	0.2223
κ	0.35	0.4998	0.2233
	0.5	0.4938	0.2194 (d)
	0.7	0.4842	0.2100
	1.0	0.4249	0.1779
λ	0.0	0.4563	0.2010
	0.1	0.4845	0.2110
	0.2	0.4938	0.2194 (d)
	0.4	0.4955	0.2208
ρ	0.0	0.4949	0.2190
	0.1	0.4952	0.2208
	0.2	0.4943	0.2199
	0.3	0.4938	0.2194 (d)
	0.5	0.4919	0.2208

E.2 Baseline Configuration

FFMSR uses BERT features and validation-based model selection. Pretraining/fine-tuning lasts 5/20 epochs for M–V and 1/5 epochs for pairs involving Books. FedDCSR uses 10 communication rounds with three local epochs per round. The sequential benchmarks contain 10,634 Music, 89,254 Books, and 18,643 Movies items. Tables 2 and 3 list all six evaluated transfers.

Table 24: Robustness (three-run base means). LLM: LLaMA features; noise results average six transfers.

Setting	HR@20	NDCG@20
5 clients	0.4940	0.2183
20 clients	0.5769	0.2788
50 clients	0.6035	0.2935
Activity split 0.1	0.4523	0.2059
Activity split 0.5	0.4839	0.2167
Popularity split 0.5	0.4895	0.2199
Fixed compute: 5 clients	0.2803	0.1089
Fixed compute: 20 clients	0.2460	0.0919
Fixed compute: 50 clients	0.2419	0.0889
Participation 0.2	0.2583	0.1034
Participation 0.5	0.2692	0.1074
Noise 0.1	0.4346	0.1938
Noise 0.2	0.4351	0.1944
Noise 0.3	0.4343	0.1941
LLM M→V	0.5091	0.2243
LLM B→V	0.4869	0.2203
LLM M→B	0.2519	0.0998

Table 25: Anchor granularity (five-run cross-domain means).

Anchoring	HR@20	NDCG@20
None	0.4346	0.1934
Catalog gate	0.4360	0.1944
Per-item gate	0.4365	0.1950

Table 26: Cosine distance from the frozen prior. Reference: TransFR.

Slice	Reference	No anchor	Base
Cross all	0.8916	0.9450	0.9841
Cross cold	0.8983	0.9721	0.9645

Fixed-compute/participation runs use eight rounds, $(\alpha, \kappa, \lambda, \rho) = (0.55, 0.7, 0.4, 0.2)$, and 150/160/150 total steps for 5/20/50 clients. Full participation uses five clients; other rows use the selected configuration.